

ATPL EASA STUDY GUIDE | 090 - COMMUNICATIONS

Chapter 1: General Procedures & Phraseology

Syllabus Source: CAE Oxford (pp. 1-32) | Pass Target: >= 90% AviationExam

1. THE ICAO STANDARD RADIOTELEPHONY ALPHABET & NUMBERS

Aeronautical radiotelephony communications require absolute clarity and zero phonetic ambiguity. Standard ICAO phonetic spelling and number pronunciation must be used across all aeronautical frequencies:

Col 1	Col 2	Col 3	Col 4	Col 5	Col 6
A - Alpha	B - Bravo	C - Charlie	D - Delta	E - Echo	F - Foxtrot
G - Golf	H - Hotel	I - India	J - Juliett	K - Kilo	L - Lima
M - Mike	N - November	O - Oscar	P - Papa	Q - Quebec	R - Romeo
S - Sierra	T - Tango	U - Uniform	V - Victor	W - Whiskey	X - X-ray
Y - Yankee	Z - Zulu	0 - ZE-RO	1 - WUN	2 - TOO	3 - TREE
4 - FOW-er	5 - FIFE	6 - SIX	7 - SEV-en	8 - AIT	9 - NIN-er

[!] EXAM TRAP (AVIATIONEXAM) - AviationExam Number Transmission Rules (ICAO Doc 4444 / Annex 10)

- General Rule: Each digit must be transmitted individually (e.g. FL 180 is 'FLIGHT LEVEL WUN AIT ZE-RO'). - EXCEPTIONS for Whole Hundreds & Thousands: - 800 is transmitted as 'AIT HUND-red'. - 5,000 is transmitted as 'FIFE TOU-SAND'. - 11,000 is transmitted as 'WUN WUN TOU-SAND'. - 38,300 is transmitted as 'TREE AIT TOU-SAND TREE HUND-red'. - Altimeter Settings: 1013 hPa is 'WUN ZE-RO WUN TREE'. 29.92 inHg is 'TOO NIN-er DAY-SEE-MAL NIN-er TOO'. - 8.33 kHz Frequencies: All 6 digits must be spoken (e.g. 118.005 is 'WUN WUN AIT DAY-SEE-MAL ZE-RO ZE-RO FIFE'), EXCEPT when the 5th and 6th digits are both zero (e.g. 118.000 is 'WUN WUN AIT DAY-SEE-MAL ZE-RO').

2. READABILITY SCALE (SIGNAL STRENGTH & CLARITY 1 TO 5)

Scale Value	Official ICAO Definition	Operational Meaning & Quality
1	Unreadable	Signal cannot be understood at all.
2	Readable now and then	Only occasional words or syllables intelligible.
3	Readable but with difficulty	Message can be understood only with severe concentration and repeats.
4	Readable	Transmission clear with minor background noise or distortion.
5	Perfectly readable	Crystal clear audio quality; zero background noise or distortion.

3. STANDARD ICAO WORDS AND PHRASES

Phrase / Word	ICAO Definition	Operational Context & Exam Nuance
ROGER	I have received all of your last transmission.	CRITICAL: ROGER does NOT mean clearance, permission, or agreement! It means receipt only.

WILCO	I understand your message and will comply with it.	Combines ROGER and compliance. Never say 'ROGER WILCO' (redundant).
AFFIRM / NEGATIVE	'Yes' / 'No'	Standard affirmations. Never use conversational slang ('Yeah', 'Nope').
STANDBY	Wait and I will call you.	The caller must pause and wait; does not grant permission to act.
CORRECTION	An error has been made in this transmission...	The correct version follows immediately ('...correction heading 280').
CLEARED	Authorized to proceed under the conditions specified.	Reserved strictly for air traffic control clearances.

4. AIRCRAFT CALLSIGNS & ABBREVIATION RULES

- Type A (Registration): Characters corresponding to aircraft registration (e.g. G-ABCD). May be abbreviated to first letter and at least last two (G-CD).
- Type B (Operator + Registration): Telephony designator of operator + registration (e.g. SPEEDBIRD G-ABCD -> SPEEDBIRD CD).
- Type C (Operator + Flight Number): Telephony designator + flight number (e.g. IBERIA 1234). NEVER ABBREVIATED under any circumstances!
- Golden Rule of Abbreviation: An aircraft station may use its abbreviated callsign ONLY AFTER it has been addressed in this manner by an air traffic communication station!

[!] EXAM TRAP (AVIATIONEXAM) - Mandatory Readback Items (AviationExam Absolute Core - ICAO Annex 11)

The flight crew MUST read back verbatim: 1. ATC route clearances; 2. Clearances to enter, land on, take off on, hold short of, cross, taxi and backtrack on ANY runway; 3. Runway-in-use, altimeter settings (QNH, QFE), SSR codes (squawk), level instructions, heading and speed instructions; 4. Transition levels.

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Algoritmo Paso a Paso (Guía para Dummies): Number Pronunciation & Readback

REGLAS MNEMOTÉCNICAS DE EXAMEN: 1. Pronunciación de números enteros: - Centenas redondas: 'HUND-red' (ej. 800 = 'AIT HUND-red'). - Millares redondos: 'TOU-SAND' (ej. 5.000 = 'FIFE TOU-SAND'). - Combinación: 38.300 = 'TREE AIT TOU-SAND TREE HUND-red'. 2. Niveles de vuelo (FL): SIEMPRE dígito a dígito (ej. FL 100 = 'FLIGHT LEVEL WUN ZE-RO ZE-RO'). 3. Escala de legibilidad: Del 1 (ininteligible) al 5 (perfecto). 'Read you 3' significa 'legible con dificultad'.

5. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: How should the frequency 118.005 MHz be transmitted over the radio under 8.33 kHz channel spacing?

- [A]: 'One eighteen point zero zero'
- [B]: 'One one eight decimal zero zero five'
- [C]: 'One one eight point zero five'
- [D]: 'One eighteen zero zero five'

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [B]. In 8.33 kHz channel spacing, all six digits must be pronounced: 'One one eight decimal zero zero five'.

Q2: How should an altimeter setting of 1013 hPa be pronounced in ICAO radiotelephony?

- [A]: 'One thousand and thirteen'
- [B]: 'Ten thirteen'
- [C]: 'One zero one three'
- [D]: 'One zero thirteen'

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CORRECT: [C]. Altimeter settings are transmitted by pronouncing each digit separately: 'One zero one three'.

Q3: What does the ICAO standard word 'ROGER' mean when transmitted by an aircraft?

- [A]: I agree with your proposal
- [B]: I have received all of your last transmission
- [C]: I will comply with your instructions
- [D]: Cleared as requested

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [B]. 'ROGER' means 'I have received all of your last transmission'. It does NOT mean compliance (which is 'WILCO') nor authorization.

Q4: When is a pilot permitted to use an abbreviated aircraft callsign (e.g. 'G-CD' instead of 'G-ABCD')?

- [A]: Anytime after initial contact
- [B]: Only after the ground station has first addressed the aircraft using the abbreviated callsign
- [C]: In emergency situations only
- [D]: When flying outside controlled airspace

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [B]. A pilot must use full callsign until the ground station initiates the abbreviation; only then may the pilot use the abbreviated callsign.

Q5: Which of the following instructions must ALWAYS be read back verbatim by the pilot?

- [A]: Meteorological wind updates
- [B]: Altimeter setting (QNH), runway-in-use, and holding point instructions
- [C]: Time checks
- [D]: Surface temperature and dew point

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [B]. Runway instructions, holding points, QNH, squawks, headings, speeds, and levels are strictly mandatory readback items under ICAO Doc 4444.

Q6: How is a transmission readability of '3' defined on the ICAO 1 to 5 scale?

- [A]: Perfectly readable
- [B]: Readable with difficulty
- [C]: Readable now and then
- [D]: Unreadable

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [B]. 1 = Unreadable, 2 = Readable now and then, 3 = Readable but with difficulty, 4 = Readable, 5 = Perfectly readable.

Q7: How should a pilot report climbing from FL 150 to FL 220 under standard ICAO radiotelephony?

- [A]: 'Climbing to two twenty'
- [B]: 'Leaving flight level one five zero, climbing to flight level two two zero'
- [C]: 'Up to flight level two two zero'
- [D]: 'Flight level two two zero now'

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CORRECT: [B]. When vacating a level, both the vacated level and the cleared level must be stated: 'Leaving flight level one five zero, climbing to flight level two two zero'.

Q8: Which component of an air traffic message does NOT require mandatory verbatim readback by the flight crew?

- [A]: Altimeter setting (QNH)
- [B]: Surface wind and ambient temperature
- [C]: Transponder SSR squawk code
- [D]: Runway holding point instructions

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CORRECT: [B]. Surface wind, temperature, and runway surface condition are advisory reports and do not require verbatim readback, unlike instructions, levels, and QNH.